Meal Delivery Supply Chain Optimization

Group A

Course: Supply Chain Management and Logistics (1001)

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1 Problem Definition

With the discovery of our meal delivery supply chain, this section clearly focuses on the core business problem, its essentials, and the specific decisions that should be made in optimization.

1.1 Three Main Problem

Our meal delivery platform operates 77 fulfillment centers and offers 51 unique meal products. With our summary, our supply chain faces a critical challenge: **how to allocate inventory across "fulfillment center-meal" combinations to balance stockout losses and overstock costs**, with adapting to three key demand dynamics which are the three specific unsolved key problems:

- *Product concentration*: A small number of meals drive most orders, but we must respect local differences across centers.
- *Center volatility*: Order fluctuations (measured by Coefficient of Variation) actually, differ by multiple times between different centers.
- *Promotion impact*: Email campaigns could increase weekly orders to some extent, but the current supply chain is lack of a mechanism to explain and respond to this surge.

1.2 Significance of Problems

The problems seriously affect the platform's longevity, having significant impacts on three core business pillars including **profitability, customer satisfaction, and operational efficiency**:

1. **Profitability**: Overstock leads to meal waste causing cost loss, while stockouts result in lost sales causing revenue loss. This poor inventory management can reduce net profit margins.
2. **Customer satisfaction**: Stockouts force customers to cancel orders or replace ours with competitors, harming brand loyalty and customer retention.
3. **Operational efficiency**: Manual replenishment needs labor hours for center managers, and some unplanned restocks increase logistics costs and shipping costs causing expedited shipping.

1.3 Business Decisions Support

Due to the issues, this analysis is especially designed and conducted to provide actionable insights for 5 critical operational decisions:

1. Inventory prioritization optimization: Which meal categories such as rice bowls, beverages and cuisines such as Indian, Thai, should be allocated more inventory, and which should be prioritized less? Identify which meal categories and SKUs to prioritize for inventory planning.
2. Replenishment mode (push vs. pull strategy): Which fulfillment centers need flexible replenishment (pull) or fixed weekly cycles (push) to avoid locational effects?

3. Promotion response: Decide if and when promotions should trigger temporary stock preparation.
4. High-priority secure: Which center-meal combinations require priority inventory to avoid stockouts?
5. Low-demand optimization: Which center-meal combinations with lower orders, can reduce inventory surplus or be removed from stock to cut waste? That is Identify low-demand center-meal combinations to avoid heavy stocking and waste.

2 Data Collection and Preparation

Preparing for conducting our main experiment analysis, this section concentrates on details of the dataset sources, data cleaning and trandformation, integration logic, filtering rules, and data limitations to ensure transparency and clarity.

2.1 Data Dictionary

All data for this analysis is sourced from the public *Food Demand Forecasting Dataset*, containing 145 weeks of historical operations for a global meal delivery platform. The dataset source is shown below:

`https://www.kaggle.com/datasets/kannanaikkal/food-demand-forecasting`

The dataset includes 5 core CSV files, each serving a specific purpose in the supply chain analysis. A detailed data dictionary for each file is provided below, including variable names, data types, descriptions, and illustrative examples. Three csv (train.csv, meal_info.csv, fulfillment_center_info.csv) are mainly used and filtered.

2.1.1 train.csv

Purpose: Stores historical order records (the primary ”fact table” for demand analysis). The detailed data dictionary can be found in *Appendix A*.

2.1.2 test.csv

Purpose: Contains future-week data (reserved for demand prediction; lacks the target variable *num_orders*). *Notice*: This data is not used in our main supply chain analysis, because it is applied for further validation or testing when we finish analysis, conclude insights and optimize supply chain models. The detailed data dictionary can be found in *Appendix B*.

2.1.3 meal_info.csv

Purpose: Provides attribute information for each meal product (a ”dimension table” linked to order data). The detailed data dictionary can be found in *Appendix C*.

2.1.4 fulfilment_center_info.csv

Purpose: Provides operational attributes for each fulfillment center (a ”dimension table” linked to order data). The detailed data dictionary can be found in *Appendix D*.

2.1.5 sample_submission.csv

Purpose: Template for submitting predicted *num_orders* for the future weeks in test.csv. The detailed data dictionary can be found in *Appendix E*.

2.2 Data Cleaning and Transformation

Two types of operations were conducted: data cleaning to fix inconsistencies and duplications, and data transformation to create and fulfill required variables for deep analysis.

2.2.1 Data Cleaning

The detailed data cleaning workflow can be found in *Appendix F*. It contains five steps from import, quality checks, column keys check, all the way to output.

2.2.2 Data Transformation

Four derived variables were created and saved in v_train_enriched table to enable deeper analysis and all calculations are reproducible: According to prior sales promotion research [1], we

Table 1: Derived Variables (Transformation Logic)

Derived Variable	Transformation Logic	Purpose
discount_rate	$\frac{\text{base_price} - \text{checkout_price}}{\text{base_price}}$	Measure actual discount intensity (0–1 range)
discount_bucket	Group *discount_rate* into 4 ranges: 0-5%, 5-10%, 10-20%, >20%	Simplify discount trend analysis
promotion_flag	1 if (*emailer_for_promotion*=1 OR *homepage_featured*=1); 0 otherwise	Combine two promotion types into a single flag
gross_sales_proxy	$checkout_price \times num_orders$	Estimate sales revenue (no real cost data available)

categorize *discount_rate* into four intervals (0-5%, 5-10%, 10-20%, >20%) to capture relationship between discount depth and consumer response, where the marginal effect of a discount is minimal below 5%, becomes more obvious in the 10–20% range, and may trigger large perceived quality concerns above 20%.

2.3 Data Removal and Keeping

To ensure data relevance and quality, the following filtering rules were applied :

- **Data kept**: All records from *train.csv* (450,000+ rows):
  1. Valid *center_id* (1–77, no out-of-range values).
  2. Valid *meal_id* (1–51, no out-of-range values).

- **Data removed**: No entire records were removed, as:
    1. The dataset had no missing values.
    2. No duplicate records were found verified via *id*, the unique record identifier.
    3. No extreme outliers were present.

2.4 Linking Tables

Our database is based on the *star schema* which was used to integrate the four files, ensuring no data redundancy and efficient querying due to our original data structure **fact table + dimension tables**. The linkage logic and visualization of our star schema are shown below:

1. **Primary key for joins**:
    - *meal_id*: Links *train.csv* (fact table) with *meal_info.csv* (dimension table).
    - *center_id*: Links *train.csv* (fact table) with *fulfilment_center_info.csv* (dimension table).
2. **Integration steps**:
    (a) Step 1: Join *train.csv* with *meal_info.csv* on *meal_id* to add meal category/cuisine.
    (b) Step 2: Join the result of Step 1 with *fulfilment_center_info.csv* on *center_id* to add center attributes.
    (c) Step 3: Export the final integrated data to *00_final_combined_table.csv* including derived deep analysis variables.

Note: *test.csv* is not used in our analysis. It is a test data or forecast data being conducted in the future. The pattern we discover in our analysis will be applied in test.csv. The detailed database star schema can be found in *Appendix G*.

2.5 Data Limitations

The dataset is sufficient for analysis but has shortages that limits analysis scope:

1. **Missing operational data**: No records of actual inventory levels, stockout events, meal waste, or procurement costs.
2. **Sales estimate only**: *gross_sales_proxy* is a revenue estimate, not actual revenue.
3. **Limited promotion data**: Promotion data is binary only, only tracks "whether a promotion was run" (0/1), lacking promotion reach, click-through rate, or promotion cost. And data is weekly-level only. Thus it cannot calculate and show promotion ROI.

3 Business Overview Analysis

In terms of business scale, this study covers weeks 1 to 145, containing a total of 456,548 observational records covering 77 fulfilment centers, 51 meal products, 14 food categories, and 4 cuisine categories. The cumulative number of orders reached 119,557,485, with an average of 26.187 orders per observation, and the sales scale agency value was approximately 3.192 billion

yuan. These results indicate that the platform has already formed a relatively mature multi-node fulfillment network and multi-category operation system. Therefore, this study faces a complex supply chain fulfillment system characterized by cross-regional, multi-center, and multi-SKU combinations.

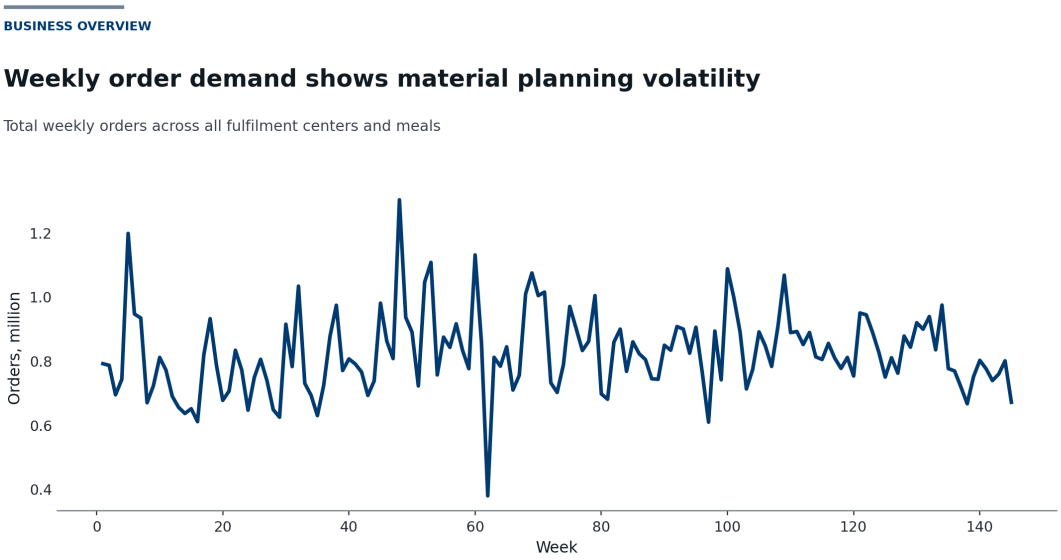


Figure 1: Weekly demand trend

Figure 1 shows the weekly order volume over 145 weeks.

From the perspective of supply chain management, this business exhibits two coexisting and mutually reinforcing basic characteristics: first, large demand volume; Second, demand is highly volatile. The larger the demand, the more significant the operational consequences of planning errors. For example, shortages of key products not only cause immediate sales losses but also weaken customer service levels; Conversely, excess stocking drives up inventory holding costs, loss costs, and pressure on end-user transfers. Especially in meal delivery scenarios, products usually have strong timeliness and short turnaround cycles, so the accuracy of demand planning directly affects procurement rhythm, capacity arrangement, warehousing and distribution coordination, and safety stock configuration. Therefore, demand management is not simply about sales forecasting, but is the core decision-making foundation that runs through procurement, inventory, distribution, and fulfillment collaboration.

From the perspective of time series characteristics, the platform's weekly demand has not remained stable but has shown significant phased fluctuations. Data shows the average weekly order volume is about 824,534 orders, but the peak weekly order volume reached 1,303,457 orders, while the off-week order volume was only 380,065 orders, resulting in a peak-to-valley ratio of about 3.43. This means that the compliance pressure the platform faces varies significantly across different operational cycles. If a company only formulates replenishment plans, production arrangements, or distribution resource allocation schemes based on long-term averages, it is prone to insufficient supply response during peak demand periods and inventory redundancy and resource idleness during demand troughs. Therefore, from the perspective of supply chain planning, demand forecasting should not remain at the static average level, but should further incorporate time trends, changes in product structure, and central heterogeneity, and build a more dynamically adaptive logic for replenishment and resource allocation.

Based on the above business overview, this report will focus on the following three core supply

chain issues:

- Which food categories and specific products constitute the main sources of demand for the platform, thereby determining key replenishment priorities and capacity assurance requirements?
- Which fulfilment centers are exposed to higher demand fluctuations and fulfillment pressures, requiring more flexible inventory and replenishment strategies?
- Have the promotion and discount mechanisms significantly altered the demand pattern and further increased uncertainty in supply chain planning and inventory control?

4 Diagnostic Analysis

This section will present three diagnostic analyses concentrating on demand management. Our overall logic follows a supply chain progress: heavily clustered product demand, erratic demand, and the promotion-activated incremental demand variability.
The aim for this sector is not only to list which variables are higher, but rather to step-by-step narrow down the core supply chain operational bottlenecks faced by the platform.

1. Product Concentration

Platforms first need to know which products consist of the large body of demand.

2. Facility-Level Variability

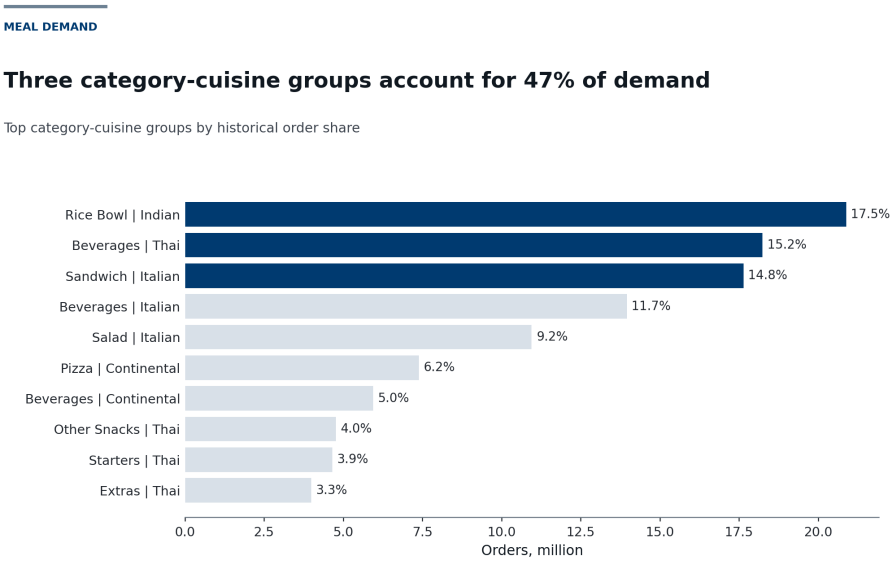
They need to understand where this demand falls and to which fulfillment centers. It qualifies demand variability across fulfillment centers to screen facilities with constantly changing demand patterns.

3. Promotional Impact

Lastly, it evaluates how promotional campaigns, including promotions and discounts, further increase short-term demand volatility.
Together, these analyses will address a key supply chain problem: which demand should be protected through stable inventory planning and which demand should be managed more flexibly.

4.1 Diagnostic 1: Meal Demand Diagnosis

Which cuisines and categories contribute most to orders? The SQL analysis first combined each category, cuisine, and SKU to calculate core operational indicators, including total orders, total meal volume, and the demand contribution ratio.
The central problem for this analysis was: if the supply chain team can only optimize a portion of their products, which categories should be prioritized? Our results show that demand is not evenly distributed; instead, it is highly concentrated in several specific categories of cuisines and meals.



The five highest-ranking combinations are:

1. Rice Bowl / Indian: 20,874,063 orders (17.46%)
2. Beverages / Thai: 18,237,630 orders (15.25%)
3. Sandwich / Italian: 17,636,782 orders (14.775%)
4. Beverages / Italian: 13,953,970 orders (11.67%)
5. Salad / Italian: 10,944,366 orders (9.15%)

Collectively, these top five product combinations contribute to nearly 70% of total platform orders. From a supply chain operational perspective, this means that inventory allocation should not be evenly distributed.
Management should be segmented into:

1. High-Demand Categories:
   (a) Ensure that the supply of raw materials is more stable
   (b) Strengthen the monitoring of the replenishment cycle
   (c) Establish a perfected out-of-stock warning system to ensure the service level.
2. Low-Demand (C-Class) Categories:
   (a) Prioritize the control of inventory holding costs
   (b) Reduce excess inventory waste
   (c) Optimize the work efficiency in the early stage of operation and stocking.

This business-level conclusion shows that this kind of product portfolio outlines a clear core business sector in the supply chain. Rice bowls, beverages, and sandwiches are not only the categories with outstanding sales performance, but also the main sources of purchasing pressure, inventory occupation and performance burden. If these core items in the core performance

center are out of stock, it will trigger the operation risk of the whole network and damage the overall performance service level of the platform, not just cause revenue losses for individual products.
Therefore, the supply chain team should regard such categories as the basic supply benchmark, rather than equally sharing efforts to control all cuisines and meals.
Further analysis at the individual SKU level confirms that there is significant demand with heavily clustered characteristics. The top-performing individual SKUs include:

1. Meal 2099 (Rice Bowl, Indian) with 9,770,303 orders
2. Meal 1885 (Beverages) with 8,346,246 orders
3. Meal 18985 (Beverages) with 8,346,246 orders
4. Meal 1745 (Sandwich, Indian) with 6,818,059 orders
5. Meal 1993 (Beverages) with 6,656,856 orders
6. Meal 11109 (Rice Bowl, Indian) with 6,377,773 orders

These popular meals are not only products with excellent sales performance, but also have a stronger influence on supply chain transmission. They cover 77 fulfillment centers and 145 weeks of continuous operational data. These core products feature full network coverage and long-cycle sales. When raw materials, kitchen process capacity, or center inventory encounter issues, the range of effect would be shown across centers.
One should establish close monitoring charts for different institutions for the final products, focusing on the high-demand items.
From the procurement perspective, high-demand meals require more stable suppliers and advanced raw material planning and buying plans. From an operational perspective, they require clearer inventory alerts at fulfillment centers. From the product strategy perspective, they are the main reason that promotion activities and campaigns may bring fulfillment pressures.
That is to say that meal demand diagnosis does not present a sales ranking list, but rather a map for prioritized management for the supply chain.
At the same time, high demand does not automatically mean that all centers should stack up their inventories. Demand diagnosis only answers which products are important and doesn't reach the part where there is high volatility. This leads directly to the next stage of analysis: center-level demand volatility.

4.2 Diagnostic 2: Center Volatility

Knowing which meals generate the highest number of orders is not sufficient for replenishment decisions, because the same meal may show different demand patterns across different fulfilment centers. Therefore, this section shifts the analysis from meal-level demand to center-level demand variability. The purpose is to identify which fulfilment centers have more volatile weekly demand and therefore require more responsive replenishment decisions.
To measure demand variability, weekly orders were first aggregated by *center_id* and week. Then, three indicators were calculated for each center: average weekly orders, standard deviation of weekly orders, and coefficient of variation. The coefficient of variation, or CV, is calculated as the standard deviation divided by average weekly orders. A higher CV indicates stronger relative demand variability.

CV is used instead of standard deviation alone because fulfilment centers may have very different business scales. A large center may naturally have a higher standard deviation, but this does not necessarily mean that it is more difficult to manage. By dividing demand variability by average orders, CV allows a fairer comparison across centers with different order volumes.

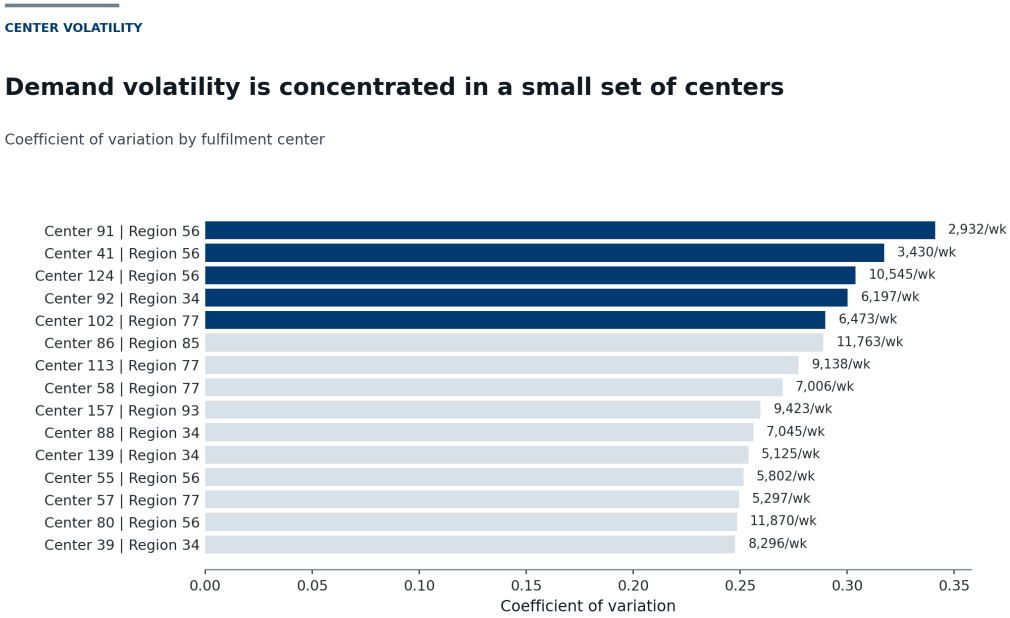


Figure 2: Coefficient of variation by fulfilment center

The results show that demand variability is concentrated in a small number of fulfilment centers. The centers with the highest demand variability include:

1. Center 91, with a CV of 0.3410 and average weekly orders of 2,932.14.
2. Center 41, with a CV of 0.3174 and average weekly orders of 3,429.92.
3. Center 124, with a CV of 0.3038 and average weekly orders of 10,544.99.
4. Center 92, with a CV of 0.3001 and average weekly orders of 6,197.33.
5. Center 102, with a CV of 0.2897 and average weekly orders of 6,472.98.

This result suggests that fulfilment centers have clear operational differences. For centers with high variability, a fixed replenishment approach creates greater risk. If inventory is planned based only on average demand, the center may face stockouts during peak weeks. However, if inventory is planned based on peak demand, low-demand weeks may lead to excess inventory and food waste. Therefore, high-variability centers are more suitable for shorter and more frequent replenishment, pre-promotion preparation, and more frequent demand monitoring.
This also shows that demand scale and demand variability represent two different inventory management problems. A high demand scale means that a stockout may affect a larger number of orders, while high variability means that demand is harder to forecast. Therefore, replenishment decisions should not rely on total order volume alone. They should also consider whether demand is stable or variable.
For practical operations, fulfilment centers can be divided into four replenishment strategy groups.

1. First, high-scale and stable centers are suitable for standardized replenishment and stable capacity planning, because they have large demand but relatively predictable demand patterns.

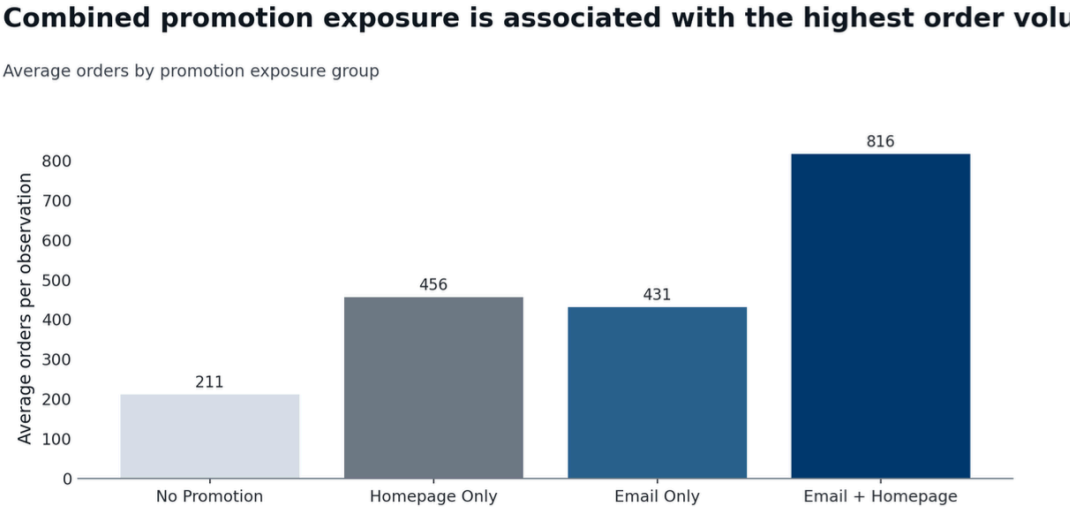
2. Second, high-scale and high-variability centers require priority inventory visibility, rolling forecasts, and pre-promotion stock checks, because they combine large business impact with high planning uncertainty.

3. Third, low-scale and high-variability centers should be managed carefully to avoid using temporary demand peaks as the basis for regular inventory levels.

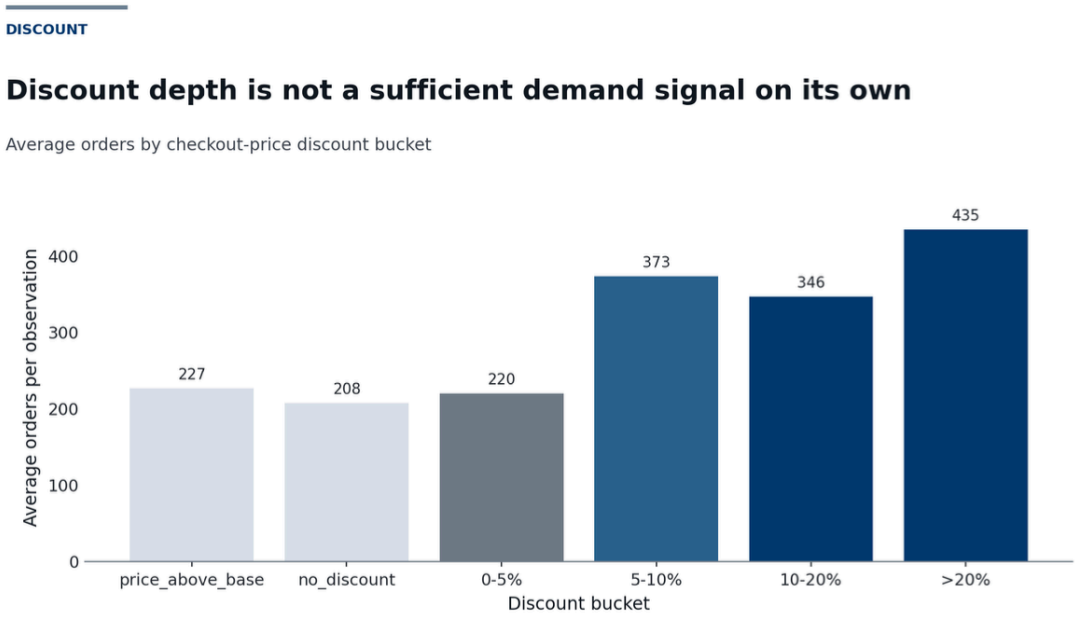
4. Fourth, low-scale and stable centers can follow basic replenishment control and standard monitoring, because their demand is relatively small and predictable.

4.3 Diagnostic 3: Promotion and Discount Diagnosis

This section aims to evaluate the correlation between promotional activities, including homepage recommendation and email marketing, and discount depth with order volume, and diagnosis impact for the supply chain management. This is significant since the replenishment strategy should meet the demand pattern in certain time period.
We first analyzed average orders per observation grouped by promotion exposure status (Figure 4).



These results reveal a strong positive relation between promotion exposure and order volume. The combination promotion exposure presents the highest order volume, and the single promotion strategies, homepage recommendation only and email only, are higher than no promotion situation.
Then, we further analyzed orders by discount bucket to isolate the effect of price discounts (Figure 5).



The results show that discount depth alone is not a sufficient demand signal. In this graph, only discount from 5-10 percent and over 20 percent produce a material demand uplift, but this is far less significance than the uplift from promotion exposure. Other discount buckets do not show a upward pattern of the order volume.

To fully understand demand drivers, we analyzed the interaction between promotion status and discount depth through a heatmap. (Figure 6)



This heatmap demonstrates that promotion status is the dominant contributor to increase of the order volume, instead of the discount depth. Even when prices are above the base price, promoted products still achieve an average of 504.31 orders, which is higher than most unpromoted discount buckets. Also, the promotion lift ratio is consistantly positive across all discount buckets. Hence, the promotion status is the primary demand signal.

As a result, the promotion is not only a marketing activity, but also a supply chain event. It would reshape the short-term demand pattern, changing the stocking pressure of the fulfillment center, and the promotion without preparation would lead to stockouts and raise the risk.

Therefore, the supply chain management team should count in the promotion status factor when planning the replenishment strategy. The promotion activity should be synchronized with the replenishment strategy. For products with high baseline demand and strong promotion response,

pre-build safety stock and secure additional raw material supply before promotions. For products with low baseline demand but high promotion dependency, adopt temporary replenishment rather than maintaining high permanent inventory levels.

5 Deep-Dive: Risk Classification + Priority Score

The previous diagonosis which are Diagnostic 1 on Meal Demand, Diagnostic 2 on Center Volatility, and Diagnostic 3 on Promotion/Discoun furnish us the basis for the decision, therefore we could bring these signals down to the `center_id + meal_id` level because replenishment is for a speific meal in a specific fulfillment center.

5.1 Analysis Granularity and Metric Design

In this section, `center_id + meal_id` os the basic unit of the analysis. Hence, this useful granularity can support us to analyze the specific meal demand levels and volatility patterns across the centers.
In order to classify each center-meal combination, we utilize the following formulas to compute the main metrics:

$$\text{Average Weekly Orders} = \frac{\text{Total Orders of a Center-Meal Combination}}{\text{Number of Active Weeks}}$$

$$\text{Coefficient of Variation (CV)} = \frac{\text{Standard Deviation of Weekly Orders}}{\text{Average Weekly Orders}}$$

$$\text{Promotion Lift Ratio} = \frac{\text{Average Orders in Promotion Weeks}}{\text{Average Orders in Non-Promotion Weeks}}$$

$$\text{Promotion Order Share} = \frac{\text{Orders in Promotion Weeks}}{\text{Total Orders}}$$

$$\text{Orders per Operating Area} = \frac{\text{Average Weekly Orders}}{\text{Fulfilment Center Operating Area}}$$

Then, we utilize the `PERCENT_RANK()` function to convert the metrics into the `demand_percentile` and `volatility_percentile` for the classification and scoring with SQL.

5.2 Demand Risk Classification

The first step is to build the demand risk classification. The classification separates center-meal combinations by demand scale, volatility, and promotion dependency. The classification logic can be found in *Appendix H*.
The four classes have different replenishment meanings. High Demand Stable combinations have high demand and relatively predictable weekly orders. High Demand Volatile combinations also have high demand, but their uncertainty is stronger. Low Demand Promotion Dependent combinations have weaker baseline demand, but their orders are more affected by promotion status. Low Demand Low Priority combinations are weak in both demand scale and promotion response.

The classification result is shown below, and the detailed output table can be found in *Appendix I*. High Demand Stable combinations contribute the largest total orders, so they are closely related to product availability and service level. High Demand Volatile combinations have similar average weekly demand, but their CV is much higher. This shows why replenishment planning should not only use average demand. Volatility needs to be kept as a separate risk dimension.

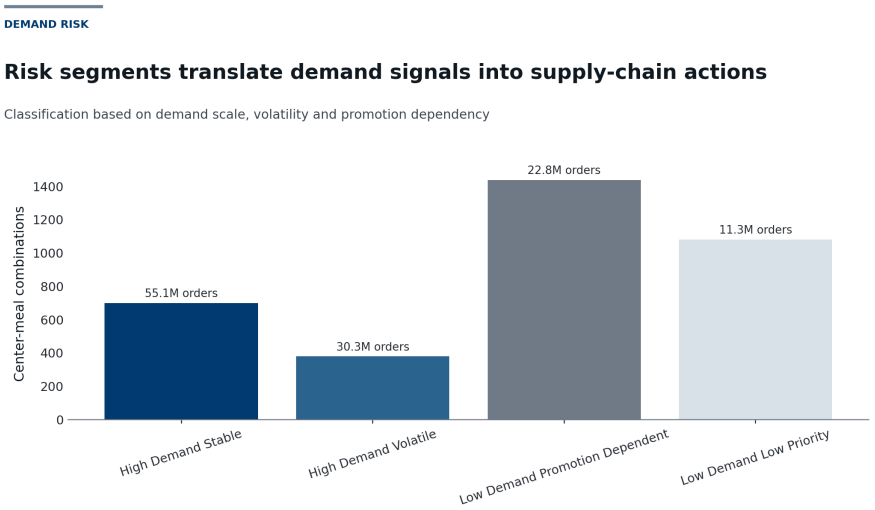


Figure 3: Demand Risk Classification Result

5.3 Demand Scale and Volatility Matrix

To further explain the classification, the following demand scale and demand volatility matrix shows that replenishment priority should not be decided only by demand scale.

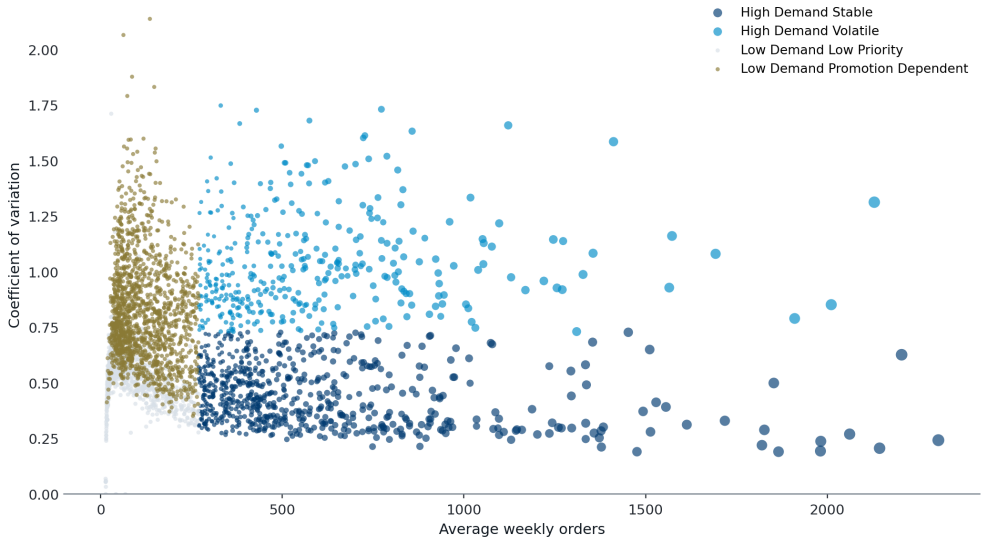


Figure 4: Demand Volatility Matrix

Combinations in the upper-right area have both high demand and high volatility. They are more likely to face stockout risk and overstock risk at the same time. If only average demand is used, their uncertainty will be underestimated. In this way, this issue illustrates the uncertainty among the model which shows that safety stock should to be considered [2].

5.4 Replenishment Priority Score

Then, to answer which Combinations Should We Manage First? We have already built a 20-point replenishment priority score using four sub-scores as follows. The score is designed to provide the attention ranking which the higher the score, the higher in the priority replenishment management. Meanwhile, each component is scored from 1 to 5 and utilizes the `PERCENT_RANK()` function as well. The detailed score conversion can be found in *Appendix J*.

$$\text{Orders per Operating Area} = \text{Average Weekly Orders} / \text{Fulfilment Center Operating Area}$$

$$\begin{aligned}\text{Replenishment Priority Score} = {} & \text{Demand Scale Score} \\ & + \text{Demand Volatility Score} \\ & + \text{Promotion Response Score} \\ & + \text{Orders per Operating Area Score}\end{aligned}$$

5.5 Key Replenishment Candidate Combinations

This figure demonstrates the highest-priority combinations and the detailed table can be found in *Appendix K*. The combinations have the common feature is that they are all in high demand scale, high volatility, string promotion response and high order density per unit operating area. In this way, they are the crucial replenishment candidates in this analysis.

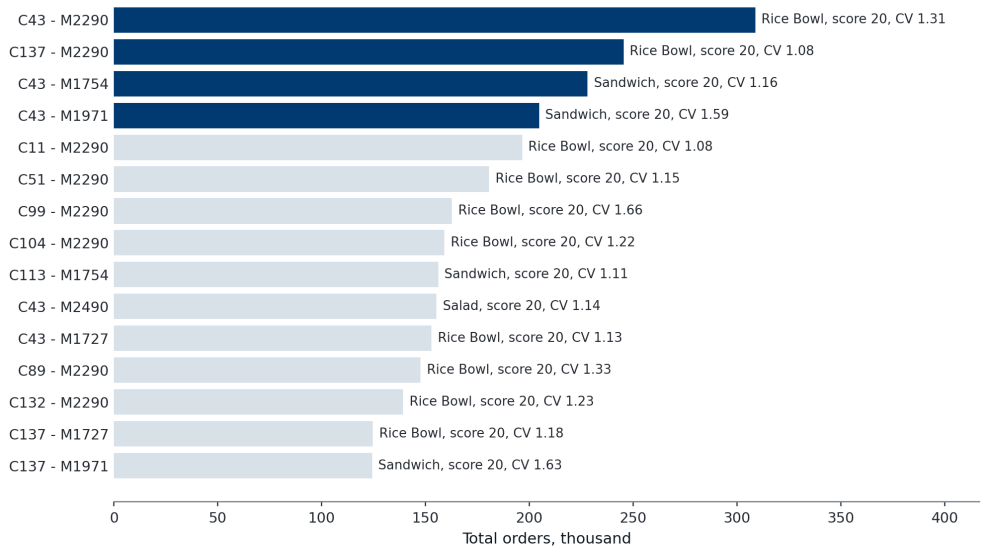


Figure 5: Replenishment Priority Score

5.6 High-Attention Centers for Flexible Replenishment

At the center level, this report counts how many High Demand Volatile combinations each fulfilment center has. The output is shown below, and the detailed table can be found in *Appendix L*. These centers need more attention because their uncertainty comes from the joint volatility of several meal combinations, not just from one unusual SKU.

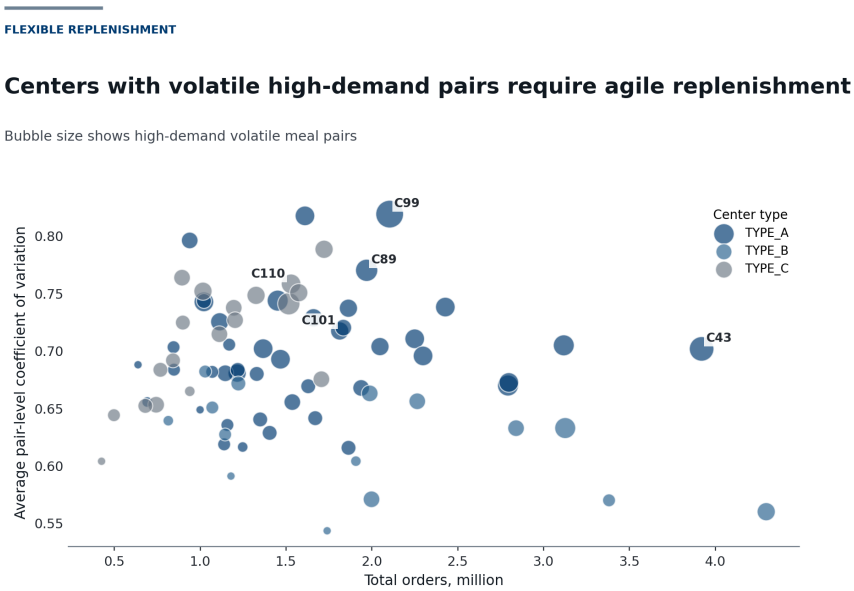


Figure 6: Flexible Replenishment Centers

In replenishment decisions, safety stock, reorder point review, and lead time all affect performance when demand is uncertain [2]. Therefore, the number of High Demand Volatile combinations at the center level can be used to identify where a more flexible replenishment mechanism is needed.

5.7 Low-Demand Low-Priority Combination Identification

The deep-dive should not only identify high-priority objects. It should also identify combinations with low demand and weak promotion response. There are 1,080 Low Demand Low Priority combinations. Their total orders are 11,251,792, their average weekly orders are only 80.83, and their average promotion lift is low.
An output sample can be found in *Appendix M*, and the detailed sample table can be found in *Appendix N*. These combinations have low demand scale and weak promotion response. Their marginal contribution to service level is limited, but they may still use inventory and operating resources. Therefore, they may create inventory investment pressure and should be separated from high-priority replenishment candidates.

5.8 Summary

This deep-dive converts diagnostic demand signals into center-meal-level inventory classification and replenishment priority. The analysis shows that replenishment decisions should consider demand scale, demand uncertainty, promotion response, and center operating pressure together. The output provides a clearer basis for the following Key Insights and Recommendations sections.

6 Key Insights

(1) The demand is highly concentrated on some kinds of food. Rice Bowl-Indian (17.5%), Beverages-Thai (15.2%) and Sandwich-Italian (14.8%) are the main parts of order, which totally account for about 47% and reflect the Pareto's rule characteristics. This means that the supply chain risks of the platform are not evenly distributed. If the key SKUs have shortage, the impact will be expanded by the size of the orders. Meanwhile. if the supply of low-demand SKUs is overstocked, the waste will be also expanded by the large number of long-tail food items.

(2) There is a significant fluctuation difference in the average demand between the fulfillment centers. The coefficient of variation for some centers is significantly higher, indicating that the fixed replenishment plan may be inadequate to cope with the real demand fluctuation.

Therefore, the platform should not only establish a single replenishment rule for all fulfilment centers. Instead, it should apply **DRP logic**, using demand forecasts, current inventory, target safety stock, replenishment quantity, and lead time to develop a time-phased replenishment plan [3].

(3) The promotion status is significantly correlated with higher order volumes. The average order amount of the "Email + Homepage" group is 816.25, which is significantly higher than the 211.42 of the "No Promotion" group.

However, when analyzing the promotional effect, we should not only pay attention to the growth ratio but also consider the original order volume of the dish. If some kinds of food usually have very few orders, even though the growth ratio is very high after the promotion, the actual increase in orders may not be large. Therefore, these kinds of food should not prepare a large quantity of stock blindly.

With regard to business aspect, this means that promotions are not only isolated marketing actions but also should be regarded as part of the supply chain process.

(4) The combination of "center_id" and "meal_id" is a more suitable management granularity for the execution of the supply chain. This means when the supply chain implements replenishment, it should not merely focus on "one food category" or "one center", but should focus on "which center and which product of that center". Therefore, summarizing data only by category or center would hide the demand risks of specific combinations.

After classifying by the "center-meal" combination, the company can determine a specific execution list, and then specify which center, which meal item, and which replenishment strategy to adopt.

(5) The priority of replenishment does not mean stocking more goods. The key of management lies in the **differentiation strategy** and **ABC analysis** logic rather than increasing inventory uniformly [2]. As the dataset does not contain actual inventory value, carrying cost, or stockout cost, this report mainly uses **demand contribution** and **replenishment priority** to identify A-class and C-class candidates:

- **High-demand and stable products** can be treated as **A-class priority items**. They should be managed through a **forecast-driven push replenishment approach**, because their demand is relatively predictable and they require stable supply assurance.

- **High-demand and fluctuant products** may also be treated as **A-class risk items**, but they should use a more **demand-driven pull approach**, supported by shorter review cycles, safety stock checks, and real-time inventory visibility.
- **Low-demand but promotion-dependent products** should use **event-driven anticipatory replenishment before promotions**, rather than maintaining high regular inventory.
- **Low-demand and low-priority products** can be treated as **C-class candidates**. They should adopt **minimum stock or on-demand replenishment** to reduce inventory carrying cost and waste.

7 Recommendations

(1) Establish a priority supply mechanism for key meals

The platform should include high-demand categories such as Rice Bowl, Beverages, and Sandwiches in the priority supply list. At the same time, the platform should also pay attention to high-score center-meal combinations, such as Center 43-Meal 2290, Center 137-Meal 2290, and Center 43-Meal 1754, etc.

- **Supporting Evidence:** Rice Bowl-Indian alone accounted for 20,874,063 orders, representing about 17.5% of total orders; Multiple Rice Bowl combinations appear on the list of top priority replenishment items, etc.
- **Execution Method:** The supply chain team can establish a ”Key Category Supply Assurance Table”, update the top center-meal combinations on a weekly basis [4], and link these combinations with the raw material purchase plan, kitchen production scheduling, and center inventory checks. For combinations that consistently maintain a high priority, the platform should set a higher service level target [5].
- **Expected Benefit:** Reducing the risk of **critical product shortages**, and enhancing the stability of fulfilling high-demand orders.
- **Risk & Limitation:** If supply management is solely based on historical orders, it may overlook new products and seasonal changes. Therefore, this list should be updated on a weekly or monthly basis.

(2) Implement DRP-based flexible replenishment planning for volatile fulfilment centers

Centers 99, 43, 89, 101, etc. should adopt a more dynamic replenishment mechanism instead of fixed-cycle replenishment. Specific measures include more frequent demand monitoring, raw material preparation before promotions, and shorter replenishment review cycles for high-fluctuation food items.

- **Supporting Evidence:** Center 99 has 14 high-demand and high-fluctuation combinations; Center 43 has 11 high-demand and high-fluctuation combinations and has a relatively large total order volume. Center 89 and 101 each have 9 high-demand and high-fluctuation combinations.
- **Execution Method:** The center can be divided into standard replenishment center and flexible replenishment center. The standard replenishment center operates in a fixed cycle; The flexible replenishment center conducts mid-week reviews, pre-promotion reviews, and abnormal order alerts. In this case, the platform can focus its management efforts on the real unstable nodes.

- **Expected Benefit:** Reducing the dual risks of stockouts during peak weeks and waste of inventory during off-peak weeks. And reducing the Bullwhip Effect [6].
- **Risk & Limitation:** Frequent restocking may increase transportation and operational costs, so it should be prioritized for high-demand and high-fluctuation combinations.

(3) Coordinate promotion planning with event-driven replenishment

Promotional emails and homepage recommendations should be synchronized with the supply chain management team in advance. For combinations with strong promotional responses, the team should conduct temporary stock preparation before the promotion; For combinations with low basic demand but high reliance on promotions, it is suitable to avoid maintaining a constant high level of inventory.

- **Supporting Evidence:** The average order amount for the "Email + Homepage" group was 816.25, which was approximately 3.86 times that of the No Promotion group. The interaction graph of promotions and discounts shows that the average order amount for multiple discount buckets under the promoted status was significantly higher.
- **Execution Method:** It is suitable to establish a supply chain review process before conducting promotions. For any homepage recommendations or email promotional activities, the historical demand, promotional response, current inventory, and feasibility of replenishment for the corresponding center-meal combinations should be confirmed before going live. For high-demand combinations, supply should be guaranteed in advance; For low-demand but promotional-sensitive combinations, short-term temporary replenishment should be adopted.
- **Expected Benefit:** Enhancing the fulfillment capability during the promotion period and avoiding the supply chain being adversely affected by marketing activities. And reducing the Bullwhip Effect [6].
- **Risk & Limitation:** There is a correlation between promotion and order volume, but this does not prove causation. Further evaluation should be conducted by considering the promotion schedule, exposure volume, and conversion rate to obtain a more accurate assessment.

(4) Control the inventory preparation for low-demand and low-priority combinations

For the low-demand and low-priority C-class candidates, the platform should maintain a low safety stock, on-demand replenishment or regional supply restriction strategy to reduce inventory waste.

- **Supporting Evidence:** The "Low Demand & Low Priority" category includes 1,080 combinations, but the total order volume is only 11,251,792, with an average weekly order of 80.83, significantly lower than that of the high-demand category.
- **Execution Method:** For constant center-meal combinations that have low priority, the regular inventory level can be reduced. And it is reasonable to conduct regular evaluations to determine whether it is necessary to implement regional removal from the shelves, reduce supply frequency, or supply only during specific promotional and seasonal windows.
- **Expected Benefit:** Releasing storage space, reducing the risk of losses, and shifting operational resources to high-value combinations.

- **Risk & Limitation:** Some low-demand products may have strategic significance or customer tail value. Therefore, before removing them from the shelves or reducing their supply, a judgment should be made based on the brand and customer experience.

8 Individual Contribution Statement

- **LIU Jiaming** was responsible for Section 1 and Section 2, covering the problem definition and data collection and preparation. The contribution included clarifying the core supply chain problems, organizing the dataset description, explaining data cleaning and transformation logic, and supporting the final project by building a clear foundation for the following analysis.

- **Wu Xin** was responsible for Section 3, Business Overview Analysis. The contribution included summarizing the overall business scale, weekly demand trend, and major operational characteristics of the platform, and supporting the final project by connecting the dataset background with the later diagnostic analysis.

- **Ma Ang** was responsible for Section 4.1, Diagnostic 1: Meal Demand Diagnosis. The contribution included analyzing meal demand concentration, identifying high-demand product categories and SKUs, and supporting the final project by providing evidence for product-level replenishment priorities.

- **Zhang Hanyuan** was responsible for Section 4.2, Diagnostic 2: Center Volatility. The contribution focused on center-level demand volatility and supported the final project by helping identify fulfilment centers that require more flexible replenishment attention.

- **Deng Songhao** was responsible for Section 4.3, Diagnostic 3: Promotion and Discount Diagnosis. The contribution included analyzing the relationship between promotion exposure, discount depth, and order volume, and supporting the final project by explaining why promotion planning should be coordinated with supply chain decisions.

- **Chen Zeming** was responsible for Section 5, Deep-Dive: Risk Classification + Priority Score. The contribution included developing the center-meal-level risk classification and replenishment priority score, and supporting the final project by translating diagnostic findings into actionable replenishment candidate lists.

- **Zhang Shuoyan** was responsible for Section 6 and Section 7, Key Insights and Recommendations. The contribution included synthesizing the major findings, converting them into management implications and recommendations, and supporting the final project by linking the analysis results to practical supply chain actions.

Appendix

Appendix A. Data Dictionary: train.csv

Variable Name	Data Type	Description	Example
id	Integer	Unique row ID for each center-meal-week observation	1379560
week	Integer	Week number of the order (historical range: 1–145)	42
center_id	Integer	Unique ID of the fulfillment center handling the order	28
meal_id	Integer	Unique ID of the meal product ordered	1885
checkout_price	Real	Final checkout price of the meal, including discount, taxes and delivery	12.99
base_price	Real	Base price of the meal	15.99
emailer_for_promotion	Integer	Flag indicating if a promotion email was sent (1=yes, 0=no)	1
homepage_featured	Integer	Flag indicating if the meal was featured on the homepage (1=yes, 0=no)	0
num_orders	Integer	Total number of orders for the ”center-meal-week” combination (core target)	356

Appendix B. Data Dictionary: test.csv

Variable Name	Data Type	Description	Example
id	Integer	Unique row ID for each center-meal-week observation	1379560
week	Integer	Week number for prediction (future range: 146–155)	148
center_id	Integer	Unique ID of the fulfillment center	43
meal_id	Integer	Unique ID of the meal product	2290
checkout_price	Real	Planned post-discount price for the future week	10.49
base_price	Real	Planned pre-discount price for the future week	12.99
emailer_for_promotion	Integer	Planned promotion email flag (1=yes, 0=no)	1
homepage_featured	Integer	Planned homepage feature flag (1=yes, 0=no)	1

Appendix C. Data Dictionary: meal_info.csv

Variable Name	Data Type	Description	Example
meal_id	Integer	Unique ID of the meal product (links to train.csv/test.csv)	1993
category	Text	Broad category of the meal (14 unique categories total)	”Rice Bowl”
cuisine	Text	Regional cuisine type of the meal (4 unique cuisines total)	”Indian”

Appendix D. Data Dictionary: fulfilment_center_info.csv

Variable Name	Data Type	Description	Example
center_id	Integer	Unique ID of the fulfillment center (links to train.csv/test.csv)	77
city_code	Integer	Anonymized code for the city where the center is located	64
region_code	Integer	Anonymized code for the region where the center is located	56
center_type	Text	Type of fulfillment center (3 types: TYPE_A/B/C)	”TYPE_B”
op_area	Real	Area of operation (in km^2)	1.8

Appendix E. Data Dictionary: sample_submission.csv

Variable Name	Data Type	Description	Example
id	Integer	Unique row ID for each center-meal-week observation	1098228
num_orders	Integer	Predicted number of orders for the future ”center-meal-week” combination	412

Appendix F. Data Cleaning Workflow

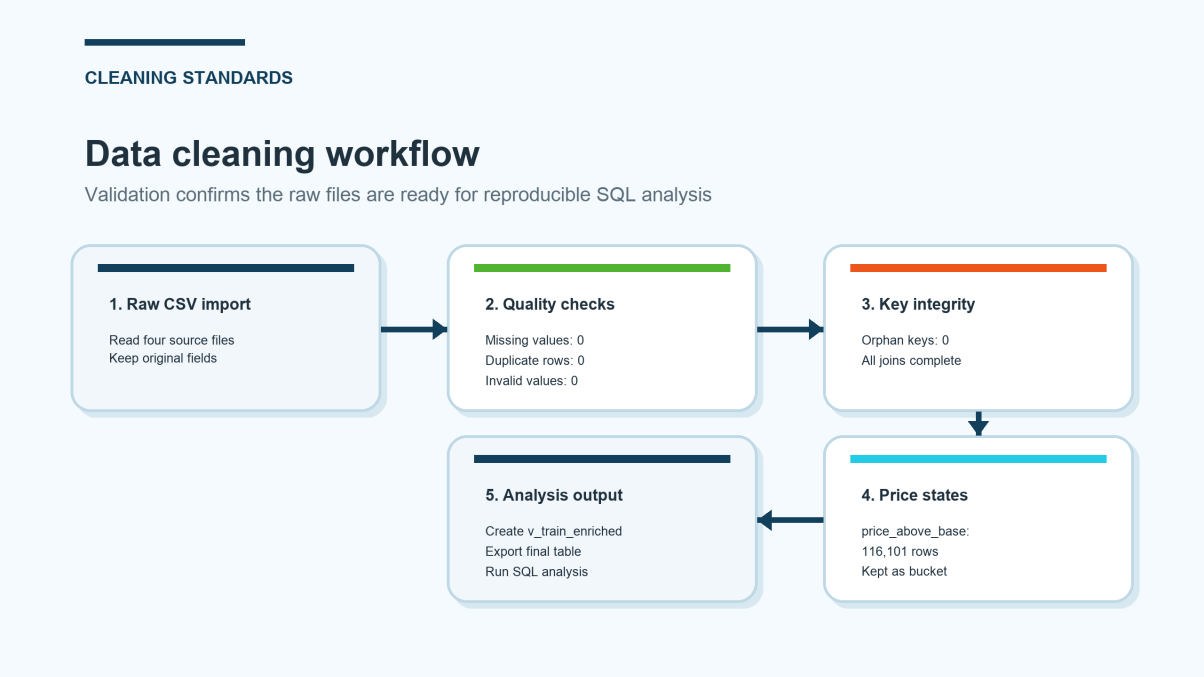


Figure 7: Data Cleaning Workflow

Appendix G. Database Star Schema

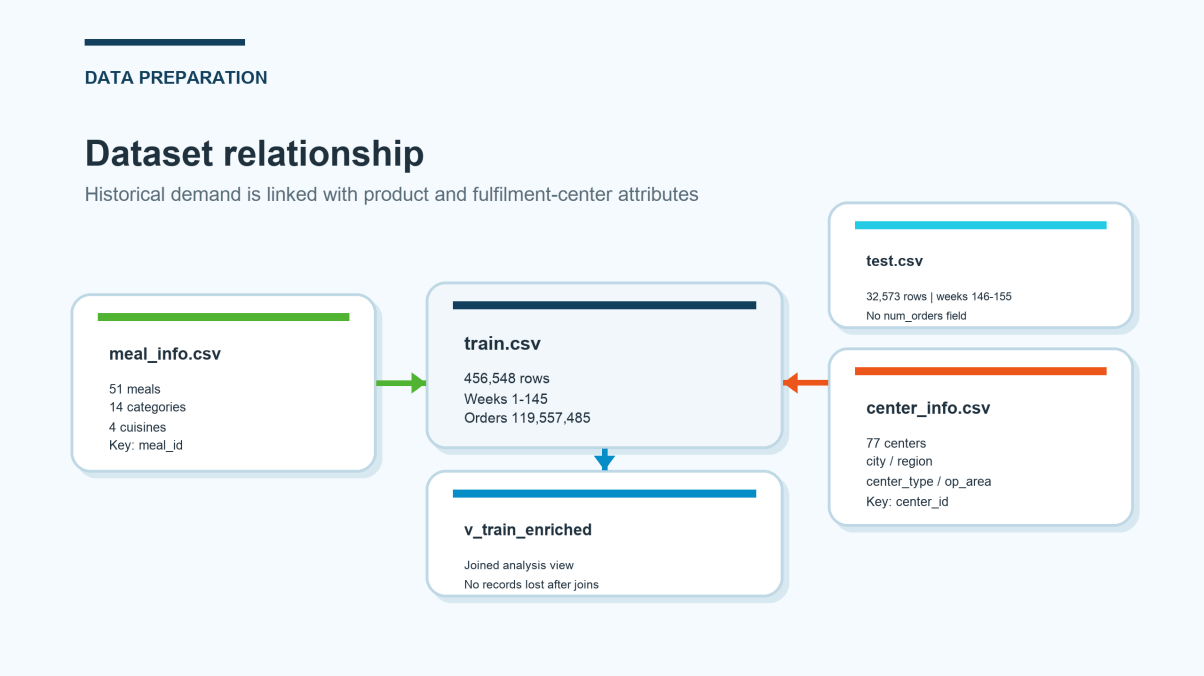


Figure 8: Database Star Schema

Appendix H. Demand Risk Classification Logic

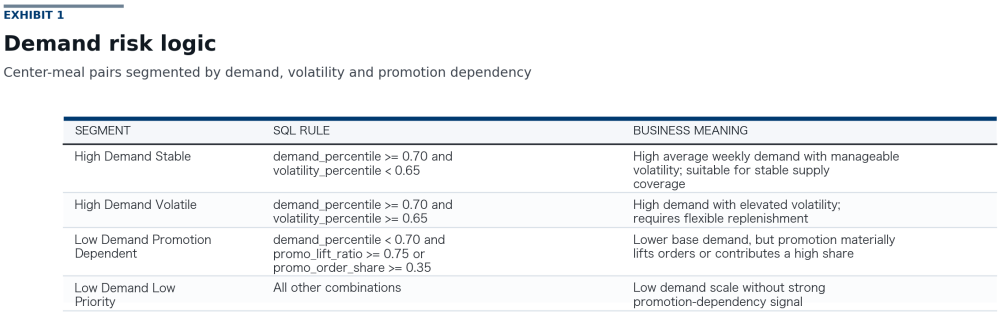


Figure 9: Demand Risk Classification Logic

Appendix I. Demand Risk Classification Summary

Table 2: Demand Risk Classification Summary

Demand risk class	Center-meal combinations	Total orders	Avg. weekly orders	Avg. CV	Avg. promo lift ratio
High Demand Stable	700	55,134,092	556.57	0.4476	0.8970
High Demand Volatile	379	30,334,033	563.94	1.0301	3.4078
Low Demand Promotion Dependent	1,438	22,837,568	119.79	0.8191	1.8901
Low Demand Low Priority	1,080	11,251,792	80.83	0.5394	0.1937

Appendix J. Percentile Score Conversion

Table 3: Percentile Score Conversion

Percentile band	Score
Top 20%	5
60%–80%	4
40%–60%	3
20%–40%	2
Bottom 20%	1

Appendix K. Top Replenishment Priority Candidate Combinations

Table 4: Top Replenishment Priority Candidate Combinations

Center	Meal	Category / Cuisine	Total orders	Avg. weekly orders	CV	Promo lift	Priority score
43	2290	Rice Bowl / Indian	308,764	2,129.41	1.3132	4.4921	20
137	2290	Rice Bowl / Indian	245,426	1,692.59	1.0809	3.9396	20
43	1754	Sandwich / Italian	228,033	1,572.64	1.1613	4.2606	20
43	1971	Sandwich / Italian	204,688	1,411.64	1.5854	7.8723	20
11	2290	Rice Bowl / Indian	196,526	1,355.35	1.0836	3.7678	20

Appendix L. Centers Requiring Higher Attention for Flexible Replenishment

Table 5: Centers Requiring Higher Attention for Flexible Replenishment

Center	Region	Type	Total orders	High demand volatile pairs	Volatile pair share
99	71	TYPE_A	2,103,282	14	28.00%
43	56	TYPE_A	3,920,294	11	21.57%
89	56	TYPE_A	1,969,200	9	17.65%
101	85	TYPE_C	1,515,771	9	18.37%
110	77	TYPE_A	1,451,046	8	17.78%

Appendix M. Low-Priority Sample Output

EXHIBIT 4

Low-priority sample

Lowest-order center-meal pairs show weak demand and limited promotion response

CENTER	MEAL	CATEGORY	CUISINE	TOTAL ORDERS	AVG WEEKLY	CV	PROMO LIFT	PROMO SHARE	DEMAND PCT
101	1571	Fish	Continental	14	14.00	0.0000	0.0000	0.0000	0.0006
139	2577	Starters	Thai	41	13.67	0.0690	0.0000	0.0000	0.0000
145	1571	Fish	Continental	42	42.00	0.0000	0.0000	0.0000	0.1527
92	2577	Starters	Thai	43	14.33	0.0329	0.0000	0.0000	0.0014
91	2494	Soup	Thai	58	14.50	0.0597	0.0000	0.0000	0.0017

Figure 10: Low-Priority Sample

Appendix N. Examples of Low-Priority Combinations

Table 6: Examples of Low-Priority Combinations

Center	Meal	Category / Cuisine	Total orders	Avg. weekly orders	Promo lift	Promo order share
101	1571	Fish / Continental	14	14.00	0.0000	0.0000
139	2577	Starters / Thai	41	13.67	0.0000	0.0000
145	1571	Fish / Continental	42	42.00	0.0000	0.0000
92	2577	Starters / Thai	43	14.33	0.0000	0.0000

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